

Session 21: AI Safety, Misuse & Governance

You've learned how AI reads, sees, hears, and combines senses. You've explored how AI is reshaping industries, creating new careers, and how bias can cause real harm. Now it's time to ask an even bigger question: **Can AI be dangerous? How can it be misused? And what rules and safeguards do we need to keep it safe for everyone?**

Today, you'll discover the dark side of AI and the bright side of human responsibility.



Objectives

01

Identify Misuse

Identify major ways AI can be deliberately misused to cause harm.

02

Define AI Safety

Define AI safety and explain why systems must be built to behave reliably.

03

Recognize Deepfakes

Recognize deepfakes and understand how they threaten trust and truth.

04

Understand Privacy

Understand privacy risks posed by AI systems that collect personal data.

05

Explore Governance

Explore governance and regulation efforts to keep AI safe and accountable.

06

Human-in-the-Loop

Apply the Human-in-the-Loop principle to safety and oversight.

Recap: AI Bias & Fairness

Before we dive into today's session, let's revisit what we covered in Session 20.

Concept	Key Takeaway
Bias	AI produces systematically unfair outcomes when trained on imbalanced data or poorly designed.
Fairness	The active, ongoing effort to ensure AI treats all groups equitably.
Representation	Diverse data and diverse teams are essential for fair AI.
Sources of Bias	Data bias, algorithm bias, and human bias often overlap.
Detection & Mitigation	Auditing data, testing across groups, and Human-in-the-Loop oversight are key.
Real-World Impact	Biased AI has denied jobs, misallocated healthcare, and rendered people invisible.

 Today we go further beyond fairness, into safety. When AI is deliberately misused or unintentionally fails, the consequences can be severe, and humans must be ready.



AI Misuse: When AI Becomes a Weapon

AI misuse happens when people **deliberately use artificial intelligence to cause harm**. The AI itself has no intentions - it is a tool, like a hammer. A hammer can build a house or break a window. The choice belongs to the person holding it.

- ① AI misuse is rarely about AI "going rogue." It is almost always about humans using AI as a tool to harm other humans. The technology **amplifies human intent** - for better or for worse.

AI Misuse



Disinformation at Scale

Generative AI can produce thousands of fake news articles, social media posts, and deepfake videos in seconds, flooding the internet with false narratives and manipulating public opinion.



Cyber Attacks

AI can help criminals generate convincing phishing emails, crack passwords, or find software vulnerabilities faster than any human hacker.



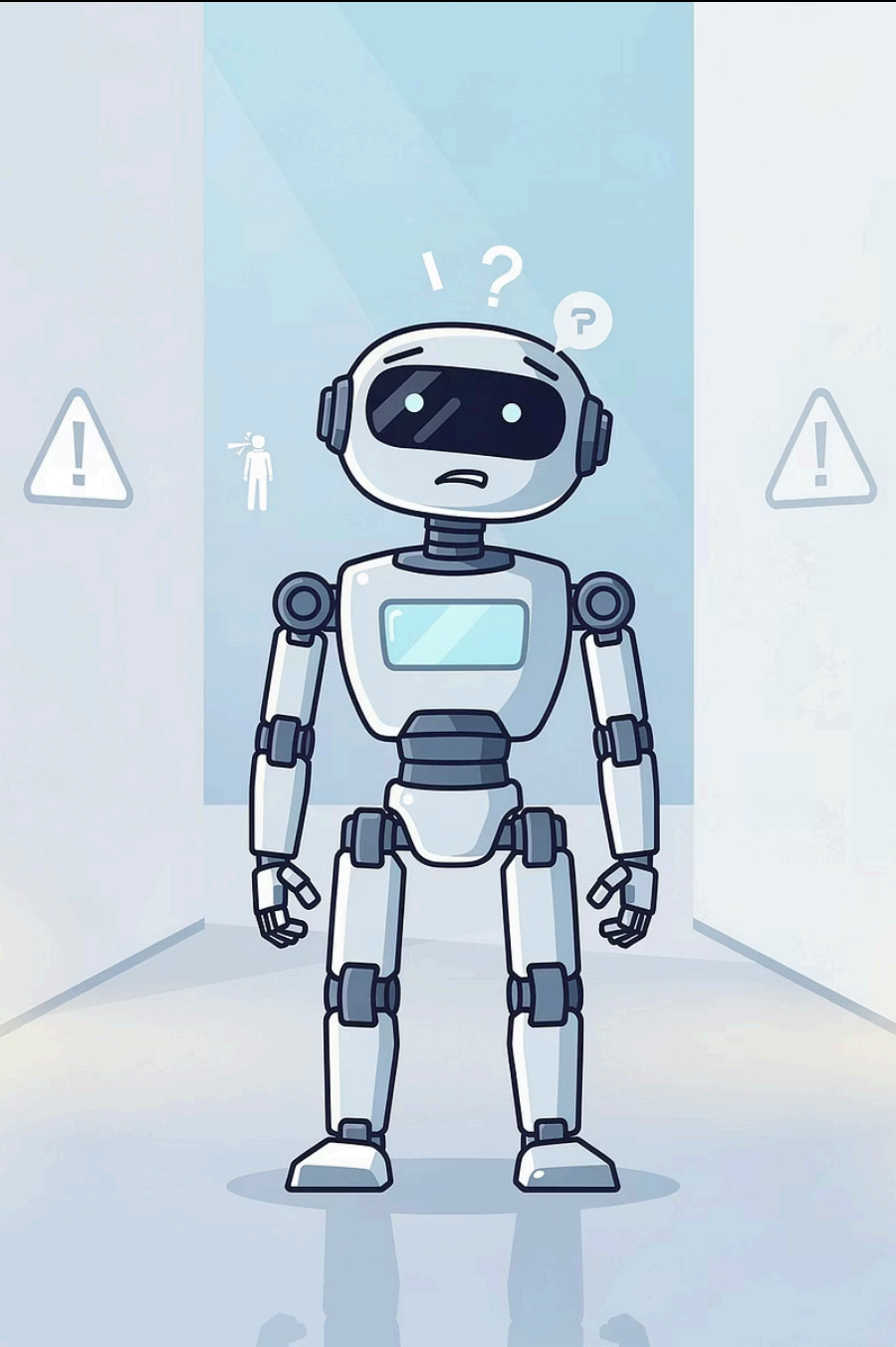
Fraud and Impersonation

Voice cloning and deepfake technology can be used to impersonate real people, tricking victims into transferring money or revealing sensitive information.



Harassment & Exploitation

AI can be used to create fake, harmful content about real people including non-consensual deepfake imagery causing severe emotional and reputational damage.



AI Safety

While AI misuse is deliberate, **AI safety concerns** are about unintended failures - situations where an AI system behaves in a way its creators did not intend, causing harm even though no one meant for it to happen.

- ⚠ AI has no intentions - good or bad. AI Safety depends entirely on how humans design, test, and oversee AI systems before they are deployed.

Examples of AI Safety Failures

Self-Driving Car

A self-driving vehicle fails to detect a pedestrian in unusual lighting conditions, resulting in a fatal collision. The system was never tested in that specific scenario.

Medical Chatbot

An AI health assistant provides dangerously incorrect medical advice to a user in crisis, because it was trained on general data and lacked clinical safeguards.

Facial Recognition

A facial recognition system misidentifies an innocent person as a criminal suspect, leading to wrongful arrest with documented higher error rates for darker skin tones.

Mini Activity 1: Misuse or Safety Failure?

Read each scenario. Decide whether the AI incident is a case of **AI Misuse** (a human deliberately used AI to cause harm) or an **AI Safety Failure** (the AI caused unintended harm due to a design flaw, bad data, or an unforeseen situation).

#	Scenario	Your Answer
1	A political campaign team uses a generative AI tool to create and distribute thousands of fake videos showing an opponent saying offensive things.	
2	A self-driving delivery robot fails to recognise a person in a wheelchair as a pedestrian because the training data contained mostly standing adults. The robot does not stop.	
3	A student uses an AI voice-cloning tool to create a fake recording of a classmate saying threatening things about the school, then shares it with the principal.	
4	An AI system used to screen rental applications starts rejecting every applicant from a specific city because a software update accidentally introduced a bug that misreads those addresses.	
5	A company trains a customer-service chatbot on its internal help documents, but the chatbot gives dangerously wrong instructions to a customer because the training documents contained errors no one had reviewed.	

Mini Activity 1: Answer Key

#	Scenario	Answer	Reasoning
1	Fake videos of opponent	AI Misuse	Humans deliberately used generative AI to deceive voters and damage a reputation.
2	Delivery robot fails on wheelchair user	AI Safety Failure	The AI was not trained on diverse enough examples - an unintended flaw, not a deliberate act.
3	Student creates fake threatening recording	AI Misuse	The AI tool was intentionally used to harass and deceive. The harm was deliberate.
4	Rental AI rejects all from a city due to bug	AI Safety Failure	An accidental software bug caused the harm. There was no human intention to discriminate.
5	Chatbot gives dangerous advice from bad docs	AI Safety Failure	The AI faithfully learned from flawed training data—an oversight in design and testing, not intentional harm.



Deepfakes: When Seeing Is No Longer Believing

A **deepfake** is a piece of media - a video, image, or audio recording—that has been artificially generated or manipulated using AI to make it look or sound like a real person said or did something they never actually did. The word comes from combining "*deep learning*" with "*fake*."

The Harms of Deepfakes



Political Manipulation

Fake videos of leaders can influence elections and destabilise democracies.



Personal Harm

Innocent people can be depicted in fake, compromising situations without consent.



Fraud

Audio deepfakes have already been used to trick companies into transferring large sums of money.



Erosion of Trust

When anyone can claim a real video is "just a deepfake," even genuine evidence becomes questionable.

⚠ Deepfakes exploit a fundamental human instinct - we tend to believe what we see and hear. In an age where AI can manufacture evidence, we must learn a new kind of **digital skepticism: verify before you trust.**

Deepfake Detection Techniques



Blink Rate Analysis

Real humans blink naturally at irregular intervals. Early deepfake models often produced faces that blinked too little or too uniformly.



Lighting & Shadow Inconsistencies

AI-generated faces sometimes have lighting that does not match the background or shadows that fall in physically impossible directions.



Heartbeat Detection

Subtle colour changes in a person's face, caused by blood flow, can be detected by sensitive cameras. Real faces have this signal; many deepfakes do not.



Audio-Visual Mismatch

The lip movements in a deepfake video may not perfectly sync with the spoken audio. Detection tools analyse the correlation between mouth shapes and sounds.

 **The Challenge:** As detection improves, so does generation. This is why human judgment, skepticism, and verification remain essential - technology alone cannot solve this problem.



Privacy Risks: Your Data in the AI Age

Modern AI systems are hungry for data. The more data they consume, the better they perform. But this creates a fundamental tension: the AI that helps you may also be **collecting, storing, and analyzing vast amounts of your personal information** - often without you fully understanding the extent.

What Kinds of Data Are at Risk?

Data Type	How AI Collects It	Potential Harm
 Voice Data	Smart speakers and voice assistants are always listening. Recordings may be stored and reviewed.	Private conversations exposed; voices cloned without consent.
 Facial Data	Facial recognition systems capture mathematical maps of your face from cameras and uploaded photos.	Your facial signature can be stolen. Unlike a password, you cannot reset your face.
 Behavioural Data	Every click, scroll, pause, and purchase is logged. AI builds profiles predicting your interests and emotions.	Your choices can be manipulated without your awareness.
 Location Data	Your phone constantly tracks your movements. AI can infer where you live, work, and socialise.	Your daily routines and private locations are exposed.
 Biometric Data	Fingerprints, iris scans, and voiceprints are used for authentication.	Once compromised, they are permanently exposed.

 **Privacy is not just about hiding things.** It is about control - the right to decide what information about you is collected, how it is used, and who can access it.

Why AI Amplifies Privacy Risks

Connecting the Dots

AI can connect dots across datasets to reveal sensitive information you never explicitly shared.

Inference Without Consent

AI can infer things about you - like your mood from your typing speed, or your health from your shopping habits.

Opacity of Systems

AI systems are often opaque: you may not know what data was collected, how it is being used, or who has access to it.

Mini Activity 2: Spot the Concern

Read each scenario. Identify whether it primarily involves a **Deepfake** concern, a **Privacy** concern, or **Both**. Be prepared to explain your reasoning.

#	Scenario	Your Answer
1	A social media account posts a video of a famous singer apparently confessing to a crime. The singer immediately denies it. Analysts find strange shadows around the singer's mouth.	
2	A smart TV manufacturer admits that their devices were recording living-room conversations even when the TV appeared to be off, and storing those recordings on company servers.	
3	A criminal uses AI to clone a child's voice from videos posted online, then calls the child's parent, mimicking the child's voice and claiming to be in trouble, demanding money.	
4	A health insurance company buys data from a fitness app that shows users' heart rates, sleep patterns, and running routes. The company uses this to adjust premiums without telling the users.	
5	A fake video of a school principal making racist remarks goes viral. The video is later proven to be AI-generated, but the principal has already been suspended and received threats.	

Mini Activity 2: Answer Key

#	Scenario	Answer	Reasoning
1	Singer confession video with artifacts	Deepfake	The AI-generated video is used to damage reputation. The primary concern is the creation of convincing fake media.
2	Smart TV secretly recording conversations	Privacy	Personal audio data was collected without consent, even when users believed the device was off.
3	AI-cloned child's voice for ransom call	Both	A voice deepfake is used for fraud (deepfake concern), and the child's voice data was scraped from online videos without consent (privacy concern).
4	Fitness app sells health data to insurers	Privacy	Sensitive personal health data was shared without user knowledge or permission.
5	Fake video of principal ruins career	Deepfake	The AI-generated video caused severe personal and professional harm. The concern is the malicious creation and spread of a deepfake.

AI Governance & Regulation

If AI is a powerful vehicle that can take us great places but also cause terrible crashes, then we need **rules of the road**. Governance and regulation are how societies create those rules.

AI governance refers to the laws, policies, ethical guidelines, and institutions that shape how AI is developed, deployed, and used.



AI Governance & Regulation



Who Can Build?

Who is allowed to build certain kinds of AI systems, and under what conditions?



What Testing Is Required?

What testing must happen before an AI system is released to the public?



Who Is Accountable?

What happens when an AI system causes harm?
Who is held responsible?



Innovation vs. Safety

How do we balance the drive for innovation with the need for safety and fairness?

AI Governance Around the World

EU AI Act

The world's first comprehensive AI law. Classifies AI applications by risk level from minimal (spam filters) to unacceptable (social scoring by governments). High-risk AI systems must meet strict requirements before deployment.

Blueprint for an AI Bill of Rights (USA)

A set of principles to guide AI policy, including the right to know when AI is being used, the right to opt out, and the right to protection from unsafe or discriminatory systems.

GDPR (EU)

A privacy law giving individuals rights over their personal data - including the right to know what data is collected, the right to have it deleted, and the right to object to automated decisions.

Global Partnership on AI (GPAI)

An international coalition of countries working together to promote responsible AI development and share best practices.

❗ Technology moves fast. Laws move slowly. The gap between what AI *can* do and what it *should* do is where governance lives. Closing that gap is one of the great global challenges.

Human-in-the-Loop: The Ultimate Safeguard

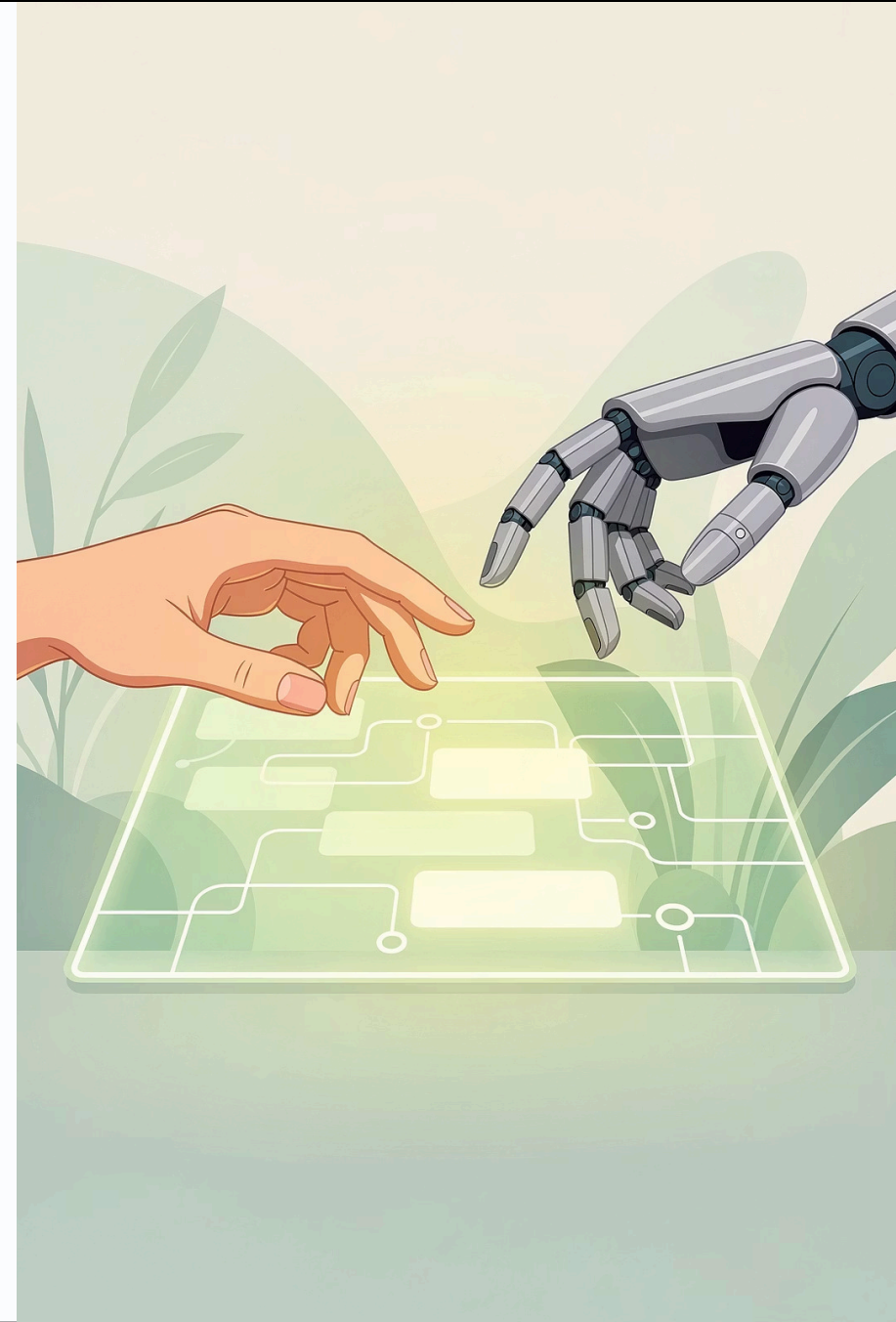
Across every concern discussed today - misuse, safety failures, deepfakes, privacy violations - one principle remains constant: **humans must stay in the loop.**

What AI Can Do

AI can analyse millions of data points per second. It can spot patterns no human would notice. It can process information at superhuman speed.

What Only Humans Can Do

AI cannot make ethical judgments. It cannot understand context the way a human can. It cannot feel the weight of a decision that affects someone's freedom, health, or life.



Hands-On Activity: Investigating an AI Safety Incident

Goal

Use NotebookLM to analyse a real-world AI safety and misuse incident, and propose governance and oversight measures.

Tool

Google NotebookLM
notebooklm.google.com

Task

You will upload a short case study, ask targeted questions, record NotebookLM's grounded answers, and then apply what you have learned today to recommend safeguards. This is the work of **AI policy advisors, compliance analysts, and Guardians**.



The Case Study: The Deepfake Phone Call

In 2019, the CEO of a UK-based energy company received a phone call from what sounded exactly like his boss - the head of the parent company in Germany. The voice on the line instructed him to urgently transfer **€220,000** (about \$243,000) to a Hungarian supplier. The CEO recognised the voice, the tone, and even the slight German accent. He authorised the transfer.

The money was gone within hours - routed through accounts in Hungary and Mexico before disappearing. The CEO later discovered his boss had never made that call. Criminals had used AI-powered voice synthesis software, trained on recordings of the executive's public speeches, to create an audio deepfake convincing enough to fool a trained professional.

This was one of the first documented cases of AI-powered voice fraud. No suspects were ever identified, and the money was never recovered.

Hands-On Activity: Steps

01

Upload to NotebookLM

Open NotebookLM and create a new notebook titled "AI Safety Case Study." Add the case study text as a source.

02

Ask Guided Questions

Use the questions on the next slide to query NotebookLM and record its grounded answers.

03

Propose Safeguards

Based on the case study and today's session, write two or three concrete measures to prevent similar incidents.

Hands-On Activity : Questions for NotebookLM

Question	NotebookLM's Answer (Based on the Source)
What AI technology was used in this incident?	
What made the deception so convincing?	
What was the immediate consequence of the incident?	
What made this case difficult to solve?	

Hands-On Activity : Propose Safeguards

Based on the case study and today's session, write two or three concrete measures the company or society - could put in place to prevent similar incidents in the future.

Your Proposed Safeguard
1.
2.
3.

Hands-On Activity: Sample Answers

Question	Sample Answer (from Source)
What AI technology was used?	AI-powered voice synthesis software - an audio deepfake.
What made the deception convincing?	The voice matched the boss's exact tone, accent, and speech patterns because it was trained on public recordings.
Immediate consequence?	€220,000 was transferred and routed through multiple countries before disappearing.
Why was it hard to solve?	No suspects were identified; the money was never recovered; audio evidence alone is difficult to trace.

Sample Safeguards

Multi-Factor Verification

Require that any financial transfer above a certain amount be confirmed through a second, independent channel - such as a video call or a pre-agreed code word.

Employee Training

Train all employees who handle finances to recognize the possibility of voice-deepfake fraud and to verify unusual requests even when they seem to come from trusted sources.

Audio Authentication Technology

Invest in or advocate for tools that can analyze audio for subtle digital artifacts that indicate AI generation.

Summary: AI Safety, Misuse & Governance

Concept	Key Idea
AI Misuse	Deliberate harmful use of AI by humans - disinformation, fraud, autonomous weapons, cyber attacks, harassment.
AI Safety	Ensuring AI systems behave as intended and do not cause unintended harm - through robust design, testing, and alignment.
Deepfake	AI-generated fake media that threatens trust, truth, and personal safety.
Privacy Risks	AI's hunger for personal data creates risks around voice, facial, behavioural, location, and biometric information.
Governance & Regulation	Laws, policies, and guidelines (like the EU AI Act and GDPR) that aim to keep AI safe, transparent, and accountable.
Human-in-the-Loop	The essential principle that humans must review and take responsibility for high-stakes AI decisions.

Homework: Safety Detection

Part 1: Spot a Concern

1

Think of an AI tool you use or have heard about. Write down one specific AI misuse, safety, or privacy concern related to that tool.

AI Tool: _____

Concern: _____

Part 2: Propose a Safeguard

2

Write one concrete action that could reduce or prevent that concern. Think like a policymaker or a Guardian.

Safeguard: _____

Part 3: Add to Your Prompt Library

3

Add this prompt to your Prompt Library:

"Explain the difference between AI misuse and an AI safety failure. Give one real-world example of each, and describe how governance or Human-in-the-Loop oversight could address the problem."

